5

The conserved-order-parameter Ising model

In the last two chapters we have looked in detail at the Ising model, which is primarily of interest as a model of a ferromagnet. In this chapter we look at a variation on the Ising model idea, the conserved-order-parameter Ising model. Although mathematically similar to the normal Ising model, the conserved-order-parameter Ising model is used to model very different systems. In particular, it can be used to study the properties of **lattice gases.**

A lattice gas is a simple model of a gas in which a large number of particles (representing atoms or molecules) move around on the vertices of a lattice. Confining the particles to a lattice in this way makes the model considerably simpler to deal with than a true gas model in which particles can take any position in space, although it also makes the model less realistic. However, lattice gases can give a good deal of insight into the general behaviour of real gases without necessarily being quantitatively accurate representations.

Many different types of lattice gases have been studied, including ones in which the particles possess inertia, or do not, ones in which particular types of particle collisions occur, and ones with a variety of different types of particle-particle interactions. In this chapter we will look at one of the simplest types which, as we will show, is equivalent to an Ising model. In our lattice gas the particles have no inertia and simply walk at random around the lattice under the influence of thermal excitations. The model is defined as follows. We start off with a lattice possessing N sites, of which a fraction ρ are occupied by particles. The particles satisfy the following rules:

1. The total number of particles is fixed, so that if a particle disappears from one site it must reappear at another. (Equivalently, we could say that the particle density ρ is constant.)

2. A lattice site can be occupied by at most one particle at any time. This **site exclusion** rule has a similar effect to the hard-sphere repulsion seen in real systems, which is the result primarily of Pauli exclusion.
3. If two particles occupy nearest-neighbour sites on the lattice, they feel an attraction with a fixed energy ϵ. This rule mimics the effect of the attractive forces between molecules.

Rules 2 and 3 are a fairly poor approximation to the forces between real molecules. For one thing, the model assumes that the repulsive and attractive forces between molecules act on the same length-scale, whereas in reality the attractive forces are normally of longer range. Nonetheless the model proves useful. This is in part because, as we mentioned above, we are often interested more in the qualitative behaviour of the system than in simulating it faithfully, but also because of the model's critical properties. As we will see, the model defined above possesses a phase transition, a critical point at which it switches from a homogeneous gas phase to a solid/vapour coexistence phase. In the vicinity of this phase transition some properties of the model become independent of the exact nature of the interactions between molecules and, simple though it is, our lattice gas is in this region capable of making quantitative predictions about real systems. This is another aspect of the phenomenon of universality which was introduced in Section 4.1.

The first step in studying our lattice gas model is to derive a Hamiltonian for it. To represent the state of the gas we define a set of variables σ_i, one on each lattice site, such that σ_i is 1 if site i is occupied by a particle and 0 if the site is vacant. We use the notation σ_i to distinguish these variables from the spins s_i of the previous chapters, which took values of ± 1.

Already, this notation ensures that Rule 2 above is obeyed. A site can only be occupied or not; we cannot have two or more particles on the same site. In order to ensure that Rule 1 is obeyed, we require that the total number of ones on the lattice, which is the number of occupied sites, should be a constant. We can express this mathematically as

$$\sum_i \sigma_i = \rho N. \tag{5.1}$$

It is Rule 3 above which specifies the Hamiltonian of our model. If two particles are located on nearest-neighbour sites, they contribute an energy $-\epsilon$ to the energy of the whole system. In terms of our particle variables we can write this as

$$H = -\epsilon \sum_{\langle ij \rangle} \sigma_i \sigma_j, \tag{5.2}$$

where $\langle ij \rangle$ denotes pairs of spins i and j which are nearest neighbours, just as in previous chapters. Now let us define a new set of variables

$$s_i = 2\sigma_i - 1. \tag{5.3}$$

These new variables are precisely the Ising spins of previous chapters. The variable s_i takes the value $+1$ on a site occupied by a particle, and -1 on an unoccupied site. Inverting (5.3) to get $\sigma_i = \frac{1}{2}(s_i + 1)$ and substituting into Equation (5.2) we then find

$$\begin{aligned} H &= -\frac{1}{4}\epsilon \sum_{\langle ij \rangle} (s_i + 1)(s_j + 1) \\ &= -\frac{1}{4}\epsilon \sum_{\langle ij \rangle} s_i s_j - \frac{1}{2} z\epsilon \sum_i s_i - \frac{1}{2} z\epsilon N, \end{aligned} \tag{5.4}$$

where z is the lattice coordination number introduced in Section 3.1 (equal to the number of nearest neighbours of each site). Making the same change of variables in Equation (5.1) we get

$$\sum_i s_i = N(2\rho - 1), \tag{5.5}$$

and substituting this into Equation (5.4) we then find that

$$H = -\frac{1}{4}\epsilon \sum_{\langle ij \rangle} s_i s_j - z\epsilon N\rho. \tag{5.6}$$

The second term is constant, since z, N and ρ all are. If we now define $J = \epsilon/4$ our Hamiltonian becomes

$$H = -J \sum_{\langle ij \rangle} s_i s_j + \text{constant}. \tag{5.7}$$

The additive constant makes no difference to the expectation value $\langle Q \rangle$ of any observable quantity Q, since it cancels out of Equation (1.7). And apart from the constant, this Hamiltonian for our lattice gas is identical to that of the Ising model, Equation (1.30), in zero magnetic field, just as we claimed above. The sum in Equation (5.5) is just the total (instantaneous) magnetization M of this model:

$$M = N(2\rho - 1). \tag{5.8}$$

(See Equation (1.33).) Since both ρ and N are constant quantities, so also is M, and for this reason we call this model the **conserved-order-parameter Ising model**, sometimes abbreviated to **COP Ising model**. (Recall that the magnetization M is also called the order parameter of the Ising model—see Section 3.7.1.) Throughout much of this chapter we will treat our lattice gas using this Ising representation. This will allow us to make use of many of the ideas we have developed in previous chapters to create algorithms for simulating the model.

Unlike the Ising model studied in previous chapters, not all states of the spins in the COP Ising model are allowed; only those for which the constraint (5.5) is satisfied are allowed. This means that the sum over states involved in calculating an expectation value (Equation (1.7)) is a sum over only these allowed states. When we come to design a Monte Carlo algorithm for this model we will need to take this constraint into account. A related point is that the COP Ising model has two free parameters we can specify. In the normal Ising model in zero magnetic field the only parameter we can vary is the temperature. In the COP Ising model on the other hand, we have to specify both the temperature and the average density of particles ρ, which in the language of the Ising model is the density of up-pointing spins or equivalently the magnetization.

From our investigations in Chapter 3 we know that in the ferromagnetic case ($J > 0$), the normal Ising model has an average magnetization which is zero above the critical temperature T_c but non-zero below (see Figure 3.9, for instance). Rearranging Equation (5.8) for ρ, this magnetization corresponds to a preferred density of

$$\rho = \frac{1}{2}\left[1 + \frac{M}{N}\right] = \tfrac{1}{2}(1+m), \tag{5.9}$$

where m is the magnetization per spin. In fact, m has two equilibrium values below T_c, one positive and one negative, so there are really two preferred densities for the model in this regime:

$$\rho_+ = \tfrac{1}{2}(1+|m|), \qquad \rho_- = \tfrac{1}{2}(1-|m|). \tag{5.10}$$

Suppose now that we fix ρ for the COP model somewhere in the range between these two preferred densities: $\rho_- \leq \rho \leq \rho_+$. The system can then arrange for the local density at every point on the lattice to take one or other of the preferred values by **phase separating** into domains of these two densities, and as we will see in Section 5.1 the COP model does exactly this. When ρ is outside the range between ρ_- and ρ_+, it is still possible for it to reach one of the preferred densities in some regions. For example, if $\rho < \rho_-$ then the system has fewer up-pointing spins than it needs to reach the preferred density, but it can still reach it in some portion of the lattice by concentrating all its up-spins in local domains. This however would leave the rest of the system starved of up-spins and even further from the preferred density than it was before. It turns out that the energetic cost of doing this is greater than the gain derived from forming the domains, so the system prefers to be homogeneous.

Thus, when $J > 0$, our COP Ising model has two phases below the critical temperature, one in which $\rho_- \leq \rho \leq \rho_+$ and there is separation into domains of high and low density, and one in which ρ lies outside this range and the system is homogeneous. The absolute magnetization $|m|$ of

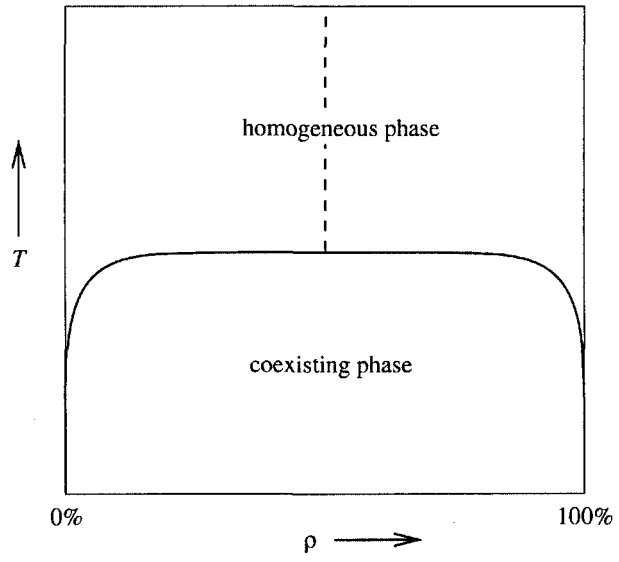


FIGURE 5.1 Phase diagram of the COP Ising model on a square lattice in two dimensions. Depending on the values of the temperature T and density ρ, the system will either be homogeneous or will separate into coexisting domains of the two preferred densities ρ_+ and ρ_-.

the normal Ising model gets smaller with increasing temperature, so that the range of ρ within which there is phase separation in the COP model, given by Equation (5.10), also gets smaller. Above T_c the range shrinks to zero and there is no phase separation. This behaviour is depicted in Figure 5.1, which shows the phase diagram for the two-dimensional COP Ising model.[1] Real fluid systems show a very similar phenomenon when they phase separate on cooling. H_2O at atmospheric pressure for example is a gas at high temperatures, but passes on cooling into a coexistence region in which it separates into distinct fractions of water vapour and liquid water. This lends support to our earlier contention that the COP Ising model shows behaviour qualitatively similar to that of real fluids.[2]

In this chapter we will show how the equilibrium properties of the COP Ising model can be efficiently simulated on a computer and discuss two ap-

[1]The line separating the two regimes, which is called the **binodal**, can be calculated exactly from the known magnetization $m = [1 - \text{cosech}^2 2\beta J]^{1/8}$ of the normal Ising model in two dimensions, which was postulated by Onsager in 1949, though not proven until Yang tackled the problem three years later (Yang 1952).

[2]Note that the high-density phase of the COP Ising model is more like a solid than a liquid however, since the "molecules" in the condensed phase of the model are arranged on the regular grid imposed by the lattice.

plications of the model to the study of physical systems. In Section 5.1.1 we will use it to examine the behaviour of phase interfaces as temperature is varied and in Section 5.3 we look at the problem of calculating the equilibrium shapes of crystals. First however, we describe the standard Monte Carlo algorithm used for simulating the model, the Kawasaki algorithm.

5.1 The Kawasaki algorithm

Neither the single-spin-flip algorithms of Chapter 3 nor the cluster algorithms of Chapter 4 can be used for the simulation of the conserved-order-parameter Ising model, because flipping either a single spin or a cluster of similarly oriented spins would change the total magnetization, which is not allowed. Instead, our Monte Carlo moves must change the values of a number of spins, chosen in such a way that the total magnetization remains the same. The simplest such algorithm is the **Kawasaki algorithm** (Kawasaki 1965), which goes as follows. We choose at random a pair of adjacent spins on the lattice. Rather than flipping these spins, our basic Monte Carlo move is to *exchange* their values. In this way the total magnetization of the lattice will always be preserved. To decide whether to exchange a given pair of spins, we calculate the change in the energy $\Delta E = E_\nu - E_\mu$ between the states μ and ν of the system before and after, and accept or reject the move with the Metropolis acceptance probability[3]

$$A(\mu \to \nu) = \begin{cases} \mathrm{e}^{-\beta \Delta E} & \text{if } \Delta E > 0 \\ 1 & \text{otherwise.} \end{cases} \tag{5.11}$$

(See Equation (3.7).)

Clearly this algorithm conserves the value of the order parameter M, but we still need to prove that it satisfies the dual criteria of ergodicity and detailed balance, so that we know that it will sample states with their correct Boltzmann probabilities. That the algorithm satisfies ergodicity can by seen most easily by considering the lattice gas representation of the model. In this representation our Monte Carlo move corresponds to the movement of a single particle from one site to another adjacent one. However, any configuration of the particles on a finite lattice can be reached in a finite number of steps from any other by moving each particle individually to its new position, and hence the algorithm is ergodic.

The demonstration of detailed balance is also very straightforward. On a lattice with N sites and lattice coordination number z, there are $\frac{1}{2}zN$ pairs

[3]In fact, Kawasaki originally proposed using the heat-bath acceptance probability, Equation (4.44), to perform the simulation. This yields a correct algorithm but, as discussed in Section 4.5.1, the Metropolis acceptance ratio is always larger than the heat-bath one and therefore gives a more efficient simulation, so most recent studies have used the Metropolis choice.

of nearest-neighbour sites. The probability of selecting any particular pair of spins at a given step of the algorithm is therefore $g(\mu \to \nu) = 2/(zN)$. The probability of selecting the same pair for the reverse move is exactly the same. The ratio of the rates for the forward and reverse moves is then required to satisfy Equation (3.3):

$$\frac{P(\mu \to \nu)}{P(\nu \to \mu)} = \frac{g(\mu \to \nu)A(\mu \to \nu)}{g(\nu \to \mu)A(\nu \to \mu)} = \frac{A(\mu \to \nu)}{A(\nu \to \mu)} = \mathrm{e}^{-\beta \Delta E}. \tag{5.12}$$

As we argued in Section 3.1, the most efficient way to satisfy this condition is the choice given in Equation (5.11).

Implementing the Kawasaki algorithm is not complicated, but there are one or two tricks worth noting. First, if we pick two spins which are pointing in the same direction then clearly there is no point in exchanging their values, since this would make no change to the state of the lattice. So it is worth checking, before we calculate ΔE, whether the spins are indeed aligned with one another. If they are, we can save ourselves some work by forgetting them and going straight on to the next Monte Carlo step.

It is tempting, in fact, to think that we might improve the speed of the algorithm by ignoring pairs of spins which are aligned and choosing only amongst those pairs which are anti-aligned. This however would be a mistake, because the number of such pairs can change when we make a move, which means that the selection probabilities $g(\mu \to \nu)$ and $g(\nu \to \mu)$ for moves in opposite directions would not be equal any more, and so would not cancel out of Equation (5.12). Choosing our spins in this fashion would result in an algorithm which does not generate states with the correct Boltzmann probabilities.[4]

In theory calculating the energy difference ΔE means calculating the energies E_μ and E_ν of the system before and after an exchange and taking their difference. As with the Metropolis algorithm however, most of the spins on the lattice are the same in states μ and ν, and we can use this fact to derive a formula for ΔE which does not involve summing over the whole lattice. The derivation follows exactly the same lines as that for the Metropolis algorithm in Section 3.1.1. We won't go through it in detail,[5] but the end result is that

$$E_\nu - E_\mu = 2J\Big[s_k^\mu \sum_{\substack{i \neq k' \text{ and} \\ \text{n.n. to } k}} s_i^\mu + s_{k'}^\mu \sum_{\substack{j \neq k \text{ and} \\ \text{n.n. to } k'}} s_j^\mu\Big]. \tag{5.13}$$

[4] Actually, it *is* possible to create a correct algorithm in this way, but one has to be quite crafty about it. And for equilibrium simulations there are, as we will shortly see, better ways than this to improve the efficiency of our algorithm. For out-of-equilibrium simulations, on the other hand, an approach which favours anti-parallel spin pairs can prove useful. This point is discussed in Section 10.3.1. (See also Problem 5.1.)

[5] The derivation appears at the end of the chapter as Problem 5.2.

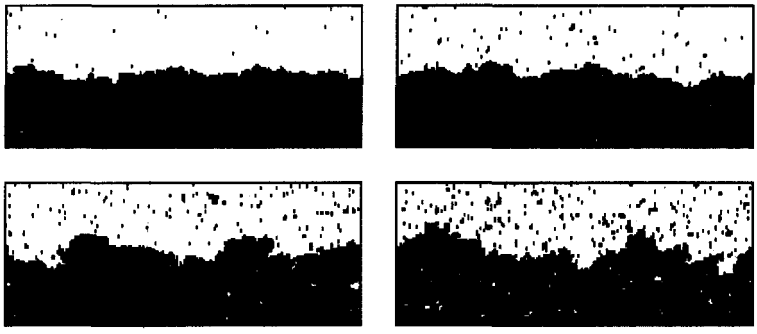


FIGURE 5.2 Typical interface configurations at several temperatures. From top left to bottom right the temperatures are T/T_c=0.6, 0.7, 0.8, and 0.9. The interface becomes fuzzier with increasing temperature and at the same time the particle densities ρ_- and ρ_+ approach each other.

Here k and k' are the two spins which we propose to exchange and the two sums are carried out over all the nearest neighbours of each spin except the two spins k and k' themselves. Notice that, like Equation (3.10) for the Metropolis algorithm, this expression only involves the values of spins in the initial state μ and not in the final state ν, which allows us to calculate the change in energy without actually having to carry out the spin exchange.

5.1.1 Simulation of interfaces

As an example of the use of the Kawasaki algorithm, we show in Figure 5.2 a set of simulations of a two-dimensional system in the phase coexistence regime in which the model separates into regions rich in particles and regions rich in vacancies. In this case, we were particularly interested in the behaviour of the interface between the two phases. In order to produce a straight interface and to prevent it from wandering we have given the system periodic boundary conditions only in the horizontal direction; at the top and bottom boundaries of the system we have fixed the spins to point down and up respectively so that the system divides into two domains with a clean interface between them.

The simulations were performed at 50% particle density (or $M = 0$ in Ising model terms) for a variety of different temperatures. As the figure shows, the interface is quite smooth at low temperatures, but becomes increasingly rough as the temperature of the simulation approaches T_c. A typical quantity of interest in such simulations is the thickness of the interface as a function of the temperature and system size. To quantify this thickness we measure the average particle density $\rho(y)$ as a function of the

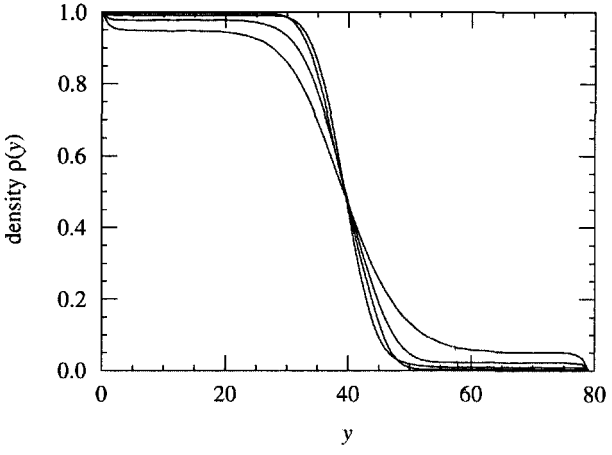


FIGURE 5.3 The average particle density $\rho(y)$ as a function of y for the four simulations shown in Figure 5.2.

coordinate y in the direction perpendicular to the interface. In Figure 5.3, for example, we show the measured values of $\rho(y)$ for the four temperatures used in Figure 5.2. A common definition of the width of the interface is the distance over which the particle density increases from $\rho_- + \frac{1}{10}\Delta\rho$ to $\rho_- + \frac{9}{10}\Delta\rho$, where $\Delta\rho = \rho_+ - \rho_-$.

In fact, the Kawasaki algorithm is not a particularly efficient way of simulating the equilibrium properties of the COP Ising model because the particles in the model move diffusively around the lattice, which is a slow way of reaching equilibrium. In the next section we develop a more efficient algorithm for the model in which particles make non-local moves.

5.2 More efficient algorithms for the COP Ising model

The speed of the Kawasaki algorithm is limited by its particular choice of Monte Carlo move. Kawasaki spin-exchange moves can only move a particle one lattice spacing at a time so that equilibration is by the diffusive motion of particles, which can be a slow process if the particles need to travel a long way. However, since we are not interested in the way in which we reach equilibrium but only in the final result, there is no need to confine ourselves to local Monte Carlo moves like this, and it turns out that we can do much better by using non-local moves. The simplest non-local move would be one in which we choose two lattice sites at random and exchange their spins,

again using a Metropolis-type acceptance ratio, Equation (5.11). Although this algorithm is actually quite a good way of simulating the COP Ising model, we can make it better still by a small change.

If we choose two spins at random there is a reasonable chance that they will be pointing in the same direction, so that exchanging their values will have no effect on the state of the system. For randomly chosen pairs the probability of their being aligned is

$$p_{\parallel} = 1 - 2\rho + 2\rho^2, \tag{5.14}$$

which has a minimum value of $\frac{1}{2}$ when $\rho = \frac{1}{2}$. Thus we are always wasting at least 50% of our moves doing nothing, and perhaps more, depending on the particle density ρ. If instead of picking pairs at random we pick one spin from the set of up-pointing spins and one from the set of down-pointing ones and exchange their values (or alternatively, flip them both over, which comes to the same thing) with a probability given by Equation (5.11), then no time is wasted on aligned pairs. This gives us an immediate improvement in efficiency by a factor of $1/(1 - p_{\parallel})$, which is at least 2 and can become arbitrarily large for ρ close to zero or one. The proof of ergodicity for this algorithm follows exactly the same lines as for the Kawasaki algorithm, as does the proof of detailed balance once we note that the selection probability for every spin pair is the same, so that the factors $g(\mu \to \nu)$ and $g(\nu \to \mu)$ cancel out of Equation (5.12) again.[6]

Implementation is a little more complex for this algorithm than it was for the Kawasaki algorithm, since in order to pick an up-spin and a down-spin we need to know where all the up- and down-spins are. Simply picking spins at random and throwing them away if they don't have the desired direction would not work—the resulting algorithm would be no more efficient than the original one which we proposed. In order to achieve the full performance gain of our new algorithm we must keep a list of all the up- and down-spins and choose our pairs of spins at random from these lists. Luckily this is quite simple to do. Since the number of up- and down-pointing spins does not change during the simulation, two arrays of fixed length can store the two lists, and when the values of two spins are exchanged, we also exchange their entries in these lists. The computational overhead incurred in doing this is relatively small,[7] and the gain in efficiency of the algorithm certainly justifies the extra work involved.

This non-local spin-exchange algorithm is already a huge improvement on the Kawasaki algorithm. For example, we find that the equilibration of

[6] Note that this algorithm obeys the condition of detailed balance perfectly well, even though, as we pointed out in Section 5.1, the equivalent local algorithm, in which we choose at random amongst only those pairs of nearest-neighbour spins which are anti-aligned, does not.

[7] See the sample program given in Appendix B.

an interface simulation like the ones depicted in Figure 5.2 requires about $\frac{1}{40}$ as many Monte Carlo steps with the new algorithm as it does with the Kawasaki one. One reason for this is the non-diffusive nature of the particle motion. Another is that the non-local update move always picks pairs of spins which are anti-aligned. The Kawasaki algorithm, as we have pointed out, sometimes picks pairs which are aligned and thereby wastes time. As it turns out, in fact, the Kawasaki algorithm picks such aligned pairs almost all of the time when we are in the phase coexistence regime, which makes it quite inefficient. A glance once more at Figure 5.2 reveals the reason for this: the Kawasaki algorithm only selects adjacent pairs of spins, but most adjacent pairs of spins are pointing in the same direction when we are in the coexistence regime. To be fair however, the non-local algorithm is more complicated to program and in our implementation we find that a single step requires about twice as much CPU time as in the Kawasaki case. Even in terms of CPU time however, the new algorithm is still a factor of 20 faster than the Kawasaki algorithm.

Our non-local algorithm is still far from perfect however. The main problem with it is the perennial one of the acceptance ratio. If the acceptance ratio is low, we are wasting a large portion of our CPU time selecting pairs and then failing to exchange their values. In the simulations of interfaces described above, for example, we find that at $T = 0.8T_c$ the average acceptance ratio for the exchange of a pair of anti-aligned spins is only about 3%.

It would be satisfying if we could find an algorithm which only considered the exchange of anti-aligned spin pairs but had a higher acceptance ratio than this, perhaps even an acceptance ratio of one. It turns out that we can achieve this goal by using the continuous time Monte Carlo method described in Section 2.4.

5.2.1 A continuous time algorithm

The basic prescription for turning our non-local algorithm into a continuous time one is to ignore all moves which do not change the state of the system (which in this case means those which involve the exchange of two spins which are pointing in the same direction) and choose between the remaining moves in proportion to their mean rates $P(\mu \to \nu)$. These rates are dictated by the condition of detailed balance:

$$\frac{P(\mu \to \nu)}{P(\nu \to \mu)} = \mathrm{e}^{-\beta(E_\nu - E_\mu)}. \tag{5.15}$$

In order to derive a general expression for the energy change $E_\nu - E_\mu$ it is helpful to define the **spin coordination number** for site i

$$n_i = \sum_{i \text{ n.n. to } j} \delta_{s_i s_j}, \tag{5.16}$$

where δ_{ij} is the Kronecker δ-symbol, which is 1 when $i = j$ and zero otherwise. The spin coordination number is the number of nearest-neighbours of site i which have spins pointing in the same direction as s_i, i.e., the number of satisfied bonds between the spin on site i and its neighbours. The change in the total energy when we update a spin at site i is then just $-2J$ times the change in the number of satisfied bonds, or $-2J(n_i^\nu - n_i^\mu)$, where n_i^μ and n_i^ν are the values of n_i in the states μ and ν. The energy change on exchanging two spins i and j is

$$E_\nu - E_\mu = -2J(n_i^\nu - n_i^\mu) - 2J(n_j^\nu - n_j^\mu). \tag{5.17}$$

Although it is not immediately obvious, this in fact is the appropriate generalization of Equation (5.13) to the case of non-local update moves. Plugging this into Equation (5.15) and rearranging a little, we get

$$\frac{P(\mu \to \nu)}{P(\nu \to \mu)} = \mathrm{e}^{2\beta J[(n_i^\nu + n_j^\nu) - (n_i^\mu + n_j^\mu)]}. \tag{5.18}$$

As we stressed in Section 2.3, detailed balance conditions such as this one only tie down the ratio of the rates for forward and backward moves, and not the rates themselves. But this equation does suggest a very simple choice for the rates:

$$P(\mu \to \nu) \propto \mathrm{e}^{-2\beta J n_i^\mu} \mathrm{e}^{-2\beta J n_j^\mu}. \tag{5.19}$$

This choice is particularly elegant because it depends only on the initial state of the lattice μ and not on the final one, and also because it factorizes into a product of two terms, one for each of the two spins involved. As we will see in a moment, this simplifies the implementation of our algorithm.

Using this choice, our continuous time Monte Carlo algorithm is now as follows. We choose an up-pointing spin i at random from all the possible choices with probability proportional to $\mathrm{e}^{-2\beta J n_i^\mu}$ and a down-pointing spin j with probability proportional to $\mathrm{e}^{-2\beta J n_j^\mu}$. The joint probability of choosing the two is then given by Equation (5.19). Then we simply exchange the values of the two spins. In order to ensure that detailed balance is preserved we must also add an amount

$$\Delta t = \frac{1}{\sum_\nu P(\mu \to \nu)} \tag{5.20}$$

to our time variable for each move we perform (see Equation (2.19)). Averages over the measured values of observable quantities then become time averages, with states weighted in proportion to the time Δt spent in them.

In practice, implementation of the algorithm involves creating separate lists of spins, divided according to whether they are pointing up or down, and by their spin coordination number. Suppose there are $m^n_{\rm up}$ members of the list of n-coordinated up-pointing spins, and $m^n_{\rm down}$ members of the list of n-coordinated down-pointing spins. Making a move involves choosing an up-pointing spin by first selecting at random one list of up-spins with probability proportional to $m^n_{\rm up}e^{-2\beta Jn}$ and then choosing one of the spins in that list uniformly at random. We also choose a down-pointing spin in the same way from the lists of down-spins and then we exchange the two. To calculate the value of Δt, we substitute Equation (5.19) into Equation (5.20) to get

$$[\Delta t]^{-1} = \sum_{i \text{ up}} e^{-2\beta J n_i^\mu} \sum_{j \text{ down}} e^{-2\beta J n_j^\mu} = \sum_q m^q_{\rm up} e^{-2\beta Jq} \sum_r m^r_{\rm down} e^{-2\beta Jr}, \tag{5.21}$$

(to within a multiplicative constant).

To compare the efficiency of this continuous time Monte Carlo algorithm with the previous two algorithms, we again turn to the simulation of an interface in two dimensions. Simulating the same systems as before, we find that the continuous time Monte Carlo algorithm requires about $\frac{1}{40}$ as many moves as our earlier non-local algorithm to bring the system to equilibrium, and about $\frac{1}{1500}$ as many moves as the Kawasaki algorithm. However, the continuous time algorithm is quite complicated to implement, and involves a lot of extra operations to keep track of the values of the sums appearing in Equation (5.21).[8] In our implementation each move of the continuous time algorithm takes about 20 times as much CPU as the simpler non-local algorithm and 40 times as much as the Kawasaki algorithm. Thus our overall gain in computational efficiency is about a factor of two compared with the non-local algorithm, and a factor of 40 compared with the Kawasaki algorithm.

5.3 Equilibrium crystal shapes

In the investigations of phase interfaces in Section 5.1.1, we kept our interfaces more or less straight by using a 50 : 50 ratio of particles to vacancies and by our choice of boundary conditions, with rows of fixed spins along the top and bottom of the lattice. These choices were appropriate for studying that particular problem, but the COP model behaves very differently if we use other densities of particles and other types of boundary conditions. For instance, on a two-dimensional lattice with periodic boundary conditions and a relatively small density of one of the two spin states, we find that

[8] Again, see the sample program given in Appendix B.

the preferred state of the model in the phase coexistence regime is one in which most of the particles in the system coalesce into a single domain or **droplet** which is usually approximately circular in shape. The reason is that the interface between domains of up- and down-pointing spins is energetically costly—the total energy of the system increases by $2J$ for every pair of anti-aligned nearest-neighbour spins along the interface. In effect, there is a **surface energy** or **surface tension** at the boundary of our droplet.[9] The system can therefore reduce its energy by reducing the length of the boundary. The lowest surface to area ratio in two dimensions is achieved when the droplet is a circle,[10] although, as we will see, it never actually takes this shape in practice because of the effects of the lattice.

Crystals display similar behaviour as they crystallize, choosing a shape which reduces their surface energy. In the limit of a large crystal this preferred shape is called the **equilibrium crystal shape** or ECS. In 1901 Wulff invented a geometrical construction for calculating equilibrium crystal shapes from a knowledge of the way in which the surface tension varies with the orientation of a surface. His construction shows that the ECS is indeed a circle (or a sphere in three dimensions) if the surface tension is isotropic. Because of the underlying crystal lattice, however, the surface tension is not isotropic and for real crystals the ECS is at best only approximately circular. Unfortunately, the orientation dependence of the surface tension is only known exactly in a very few cases, so that it is usually not possible to apply the Wulff construction to calculate the ECS. Instead we turn to Monte Carlo methods.

In order to perform a truly accurate calculation of an equilibrium crystal shape we would need to simulate in detail the interactions between the molecules in a solidifying solid. This however is not an easy thing to do. So instead many people have turned to the conserved-order-parameter Ising model as a simple model of the processes taking place. Like the real physical system, the COP Ising model shows "crystallization" in which particles coalesce out of the "melt" onto a regular crystalline lattice, and it possesses a surface tension which is orientation-dependent, giving rise to non-trivial equilibrium crystal shapes. It is true that the details of the interactions between the particles in our Ising model are very different from those in the true crystal, so we do not expect the model to give an accurate quantitative prediction of a real ECS, but, as with our earlier simulations of interfaces, the study of the COP Ising model can nonetheless shed light on the sort of processes taking place when a crystal solidifies.

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[9] For simplicity we will use the three-dimensional term "surface tension" to refer to this phenomenon in both two and three dimensions, although in two dimensions it is strictly more correct to call it "line tension".

[10] Strictly, this is only true if the density ρ of particles is sufficiently low or sufficiently high. See Problem 5.3 for an exploration of this point.

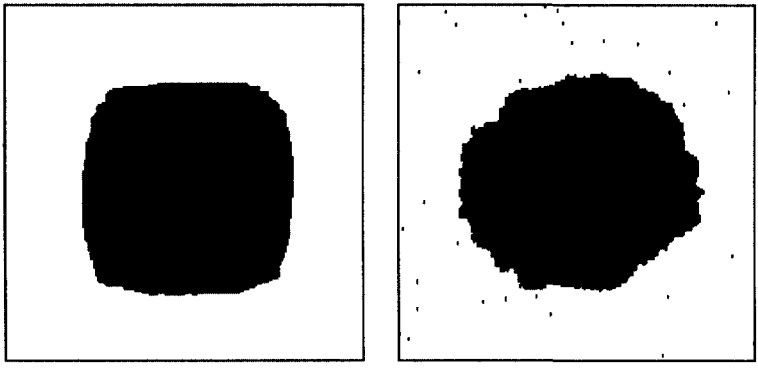


FIGURE 5.4 Typical droplet configurations at $T/T_c = 0.25$ (left) and $T/T_c = 0.5$ (right). At low temperatures the boundary of the droplet is quite sharp and the droplet's shape resembles a square. With increasing temperature the boundary becomes fuzzier and the droplet's shape becomes round.

In order to carry out a Monte Carlo simulation of an equilibrium crystal shape we could start off by placing a certain fraction ρ of particles (i.e., up-pointing spins) at random on a lattice and allowing them to move about under the action of any one of the algorithms described in the earlier sections of this chapter at a temperature T below T_c, until they coalesce into a single domain. It turns out however, that it is a lot quicker to start them off in a single domain (say a square on a square lattice) and "warm them up" to temperature T. In Figure 5.4 we show two different droplet configurations from simulations of this kind for a two-dimensional COP Ising model[11] at two different temperatures below T_c. As we can see, the droplet remains rather square at low temperatures, but becomes rounder as T increases.

To get a more quantitative estimate of the ECS, we can run our simulation for a longer time, record the shape of the droplet at intervals and then take an average over the shapes found. Unfortunately, even if we start the droplet at the centre of the lattice, it has a tendency to drift away during the simulation, so in order to average the shapes correctly we translate them first so that their centres of mass coincide.[12] The average itself simply in-

[11] We illustrate the method using a two-dimensional system for clarity. At the end of this chapter we also give some results for three-dimensional systems. However, the two-dimensional results are not purely academic. In fact, the two-dimensional COP Ising model is quite a good approximation to the behaviour of adsorbed atoms on the surfaces of metals, and in certain temperature regimes one does indeed find droplets or **islands** of adatoms which look very similar to the ones depicted in Figure 5.4. This point is discussed further in Chapter 11.

[12] In fact, our system has four-fold rotational symmetry and inversion symmetry on the

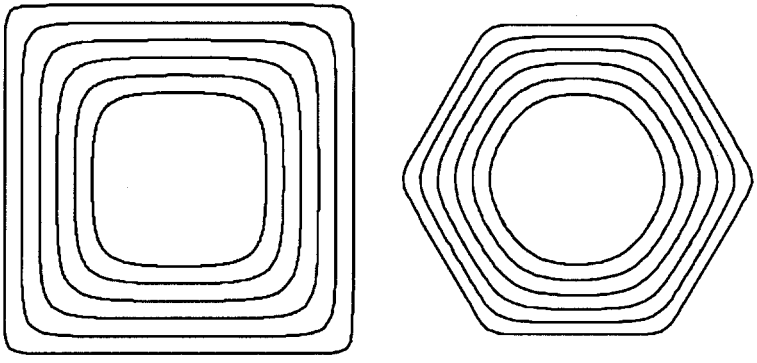


FIGURE 5.5 Equilibrium crystal shapes as a function of temperature on a square lattice (left) and a hexagonal one (right). The outermost droplet shapes correspond to inverse temperatures of $\beta J = 9.3$ in each case. Going inwards, the values of β for the other shapes are factors of 2, 3, 4, 5 and 6 lower. To make the figure clearer we have rescaled the overall size of the droplets so that the different curves do not overlap.

volves taking the mean of the particle density over all our measurements as a function of position. At the centre of mass itself this mean takes a value close to ρ_+ (see Equation (5.10)) and it remains roughly constant until we approach the droplet interface, at which point it falls quickly towards ρ_-. We define the edge of the droplet to be the point at which the density is equal to $\frac{1}{2}$.

As an example of this kind of calculation we show in Figure 5.5 measurements of the average ECS of droplets on square and triangular lattices in two dimensions. In these calculations we used the continuous time Monte Carlo algorithm of Section 5.2.1, which again proves significantly faster than either the Kawasaki algorithm or the non-local algorithm we introduced in Section 5.2.

We can also use the Monte Carlo method to study the ECS of three-dimensional systems. (Real crystals are after all three-dimensional.) The behaviour of the ECS is richer in three dimensions than in two. In two dimensions the sides of the droplet are flat at zero temperature, but become rounded for all finite temperatures. In three dimensions the story is more complicated. On a cubic lattice for instance, the ECS is a cube at $T = 0$, just as we would expect. As we raise the temperature above zero, the corners and edges of the cube become rounded, but the droplet still has flat facets.

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square lattice (the symmetry group is C_4), so we can reduce the errors on our average by using each droplet configuration eight times over in each of its symmetry equivalent orientations.

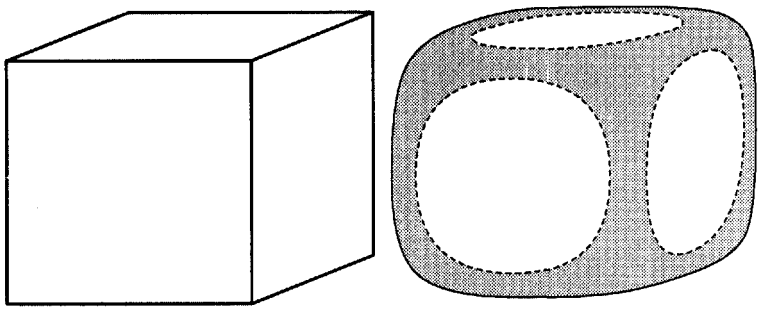


FIGURE 5.6 Schematic depiction of the equilibrium crystal shape for the COP Ising model on a cubic lattice at zero temperature (left) and at a non-zero temperature below the roughening temperature. Above the roughening temperature, the flat facets disappear.

Only when we reach a certain higher temperature, the so-called **roughening temperature** T_R, which is approximately half the critical temperature, do these facets disappear. This disappearance is called the **roughening transition** of the COP Ising model. In Figure 5.6 we give a schematic drawing of the ECS of a cubic system at zero temperature and at non-zero temperature below the roughening temperature. Above the roughening temperature, the droplet looks like a deformed sphere but has no flat surfaces.

The behaviour of equilibrium crystal shapes becomes richer still on more complex lattices and with the introduction of next-nearest-neighbour interactions. For example, Jayaprakash and Saam (1984) have studied the ECS on a face-centred cubic (fcc) lattice using a COP Ising model with both nearest- and next-nearest-neighbour interactions. In Figure 5.7 we show the ECS of the fcc crystal at zero temperature with only nearest-neighbour ferromagnetic interactions, and the ECS when small ferromagnetic next-nearest-neighbour interactions are also included. In the absence of next-nearest-neighbour interactions, the sharp corners of the ECS disappear as soon as the temperature becomes non-zero, but the facets persist up to a roughening temperature T_R, just as they do in the cubic case. Again T_R is about half the critical temperature. In the presence of ferromagnetic next-nearest-neighbour interactions, extra (110) facets appear which have a roughening temperature different from that of the (111) facets. With anti-ferromagnetic next-nearest-neighbour interactions, the corners disappear for all $T > 0$, but the edges do not disappear until we reach the melting point T_c.

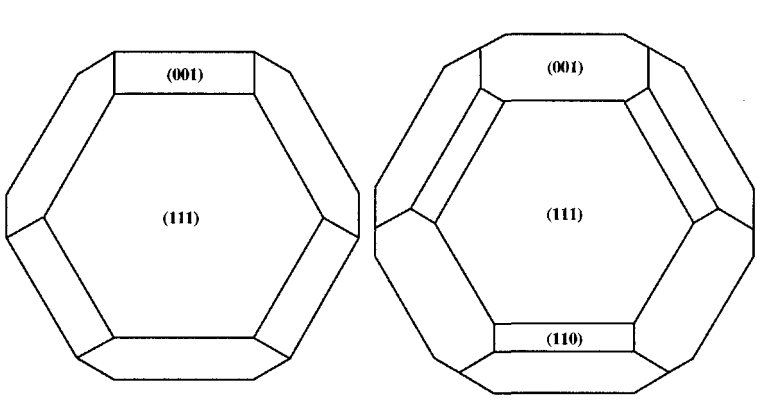


FIGURE 5.7 Equilibrium crystal shape of the COP Ising model on a face-centred cubic lattice at zero temperature without next-nearest-neighbour interactions (left) and after the introduction of small ferromagnetic next-nearest-neighbour interactions (right). After Jayaprakash and Saam (1984).

Problems

5.1 As we pointed out in Section 5.1, the variant of the Kawasaki algorithm in which we choose only among the anti-aligned nearest-neighbour spin pairs does not sample the Boltzmann distribution correctly. There is however a correct generalization of the Kawasaki algorithm which favours anti-aligned pairs and is more efficient than the standard algorithm when the density of particles ρ is far from $\frac{1}{2}$. In this variant, we choose at random a spin of the *minority* type (up or down, particle or vacancy) and then choose at random one of its neighbours, and exchange the values of the pair with the Metropolis acceptance ratio, Equation (5.11). Prove that this algorithm satisfies the conditions of ergodicity and detailed balance. Why is it more efficient than the standard Kawasaki algorithm?

5.2 Derive Equation (5.13).

5.3 Consider a COP Ising system on an $L \times L$ square lattice in two dimensions with a density of particles ρ. Calculate the total length of interface in the system (in terms of number of unsatisfied bonds) if (a) the particles coalesce into a single droplet and (b) they form a band with straight interfaces which stretches all the way across the system and wraps around the boundary conditions. Hence show that the droplet is not the preferred configuration of the system below T_c when $\frac{1}{4} < \rho < \frac{3}{4}$.